

Thomas Kopalidis

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ABOUT ME

Data Scientist & AI Engineer with 4+ years of experience building production Machine Learning, Deep Learning, and Generative AI systems across healthcare, mobility, and real estate, with additional experience as a Data Analyst in hospitality. Published researcher (Highly Cited article, MDPI Information). Skilled in Python, SQL, NLP/LLMs, LangGraph, and full-stack ML pipelines.

Technical Skills

- **GenAI & LLMs:** LangChain, LandGraph, Hugging Face, Prompt Engineering, LLM Integration, RAG.
- **Programming & Data:** Python (PyTorch, Keras, Scikit-Learn, NumPy, Pandas), SQL, R.
- **Cloud:** Azure (Databricks), AWS
- **Visualization:** Power Bi, Matplotlib, Seaborn
- **Tools:** Git, Docker, CI/CD, Flask, Postman, PostgreSQL, LaTeX, APIs

WORK EXPERIENCE

AI Engineer Consultant

PwC Greece

GR (April 2026 – May 2026)

- Developed a production-ready Agentic RAG system for internal policy Q&A. Built a LangGraph-orchestrated REST API enabling multi-turn, context-aware responses over versioned policy documents (PDF), with step-by-step agent reasoning, dynamic tool selection, and evidence-grounded answers. Containerized the full stack with Docker; applied enterprise best practices including session memory management and scalable document ingestion. Version-controlled via Git.

Operational Data Analyst & Financial Reporting

Elysium

Rhodes, GR (May 2025 - Nov. 2025)

- Designed executive dashboards (Excel/BI) to visualize Year over Year trends and demographics (e.g., top markets: Germany 27%, UK 15%), translating KPIs into revenue strategies.
- Managed daily financial reconciliation for 300+ rooms via Fidelio Opera PMS, analyzing peak traffic days (Sundays) to optimize staffing and resource allocation.

Data Scientist & AI Engineer

CERTH ITI

Thessaloniki, GR (Feb 2022 - Feb 2025)

- **Medical AI (Heart Failure Project):** Built an NLP-driven Clinical Decision Support System (CDSS) that automated MAGGIC risk scoring from unstructured clinical notes reducing manual assessment time by ~40%. Integrated LLMs for patient-facing explanations; validated across 50 clinical cases with cardiologist oversight.
- **Neurodegenerative Disease Health:** Designed a lightweight Custom CNN (3 Conv Layers, 1024 Dense Units) using Keras/TensorFlow for Facial Expression Recognition (FER). Achieved 62% accuracy on the FER-2013 dataset, outperforming SOTA VGG-Face models in efficiency for specific demographics.
- **Smart Mobility:** Architected a full-stack pipeline (Flask, PostgreSQL) for user mobility tracking. Implemented SQL indexing strategies that reduced data retrieval time by 30%. Developed a Random Forest transport classifier (85 % accuracy across 5 modes such as bus, car, walking, micromobility, multimodal) used for municipal planning. Co-developed the [HIT application](#) backend.
- **Real Estate Predictive Modeling:** Built an ML pricing pipeline for 10,000+ property records. Conducted rigorous EDA and feature selection to mitigate multicollinearity and achieved an R^2 of 0.80 using optimized Random Forest Regressors.
- **Environmental Monitoring & Disaster:** Processed multi-modal sensor data (LiDAR, RGB, thermal) to create robust training datasets, directly enabling the deployment of AI models for environmental hazard detection.

Medical Physics and Data Analysis Intern

AHEPA University of Hospital of Thessaloniki

Thessaloniki, GR (June 2019 - Sep 2019)

Tutor

Thessaloniki, GR (Sep 2018 – Present)

Teaching Physics, Mathematics, Biology, Chemistry and Informatics to adolescents of Secondary School, High School, and Anatolia College. Undergraduate-level Biomechanics to university student.

RESEARCH & PROJECTS

AI & DL Group

Athens, GR (Sep. 2025 – Present)

- **Markdown Document Preprocessor:** Full-stack web app (Python/Flask + JS) converting 12+ file formats to Markdown — as a preprocessing layer for LLM/RAG pipelines to reduce token overhead ([Markdown](#)).
- **Medical PII Detector:** An AI-powered tool for automatic detection of sensitive personal data (PII) in medical texts, designed to support GDPR compliance and patient data protection. Analyzes clinical notes and automatically identifies patient names, dates of birth, social security numbers, tax IDs, phone numbers, emails, addresses, and medical record IDs — highlighting each with color-coded visual labels.

- **Multilingual NER Extractor:** A client-side web application leveraging Llama 3.3 70B and Llama 4 Scout Vision AI to extract structured entities (e.g., names, locations) from text and images via the Groq API.1.
- **Meeting Summarizer:** An automated pipeline utilizing OpenAI's Whisper for speech-to-text transcription and Facebook's BART-large-CNN for abstractive summarization of diverse audio/video formats.
- **Euro-Demographics Dashboard (*Data Analysis & Interactive Web App*):** Multi-layered demographic analysis platform (Python, Streamlit, Plotly) integrating live Eurostat REST APIs. Implemented a Random Forest forecasting model for recursive population projection to 2035, K-Means clustering for interactive demographic profiling, and incorporated a context-aware LLM agent (LLaMA 3.3 70B) for natural language data querying ([Euro Demographics](#))
- **CRM & Customer Retention (Bank Churn Prediction):** Developed an end-to-end machine learning system using XGBoost and CatBoost to identify at-risk customers, directly generating actionable insights for CRM strategies. Deployed the solution as a scalable web API on Hugging Face. ([Bank churn API](#)).
- **Chest X-Ray Pneumonia Detection:** Engineered a dual-approach diagnostic framework benchmarking Deep Learning (ResNet/UNET) against Classical ML (GLCM/LBP) with 9 different ML models. Achieved top-tier performance with a LightGBM model (95.2% F1-score, 0.97 AUC). Deployed as a web app on Hugging Face ([Pneumonia API](#)).

EDUCATION

Master's in AI & Deep Learning (90 ECTS)

Athens, GR (2025 - 2026)

University of West Attica - English-taught program - Electrical & Electronics Engineering

- Specialization: Minor in Cognitive Systems
- Course: NVIDIA certification Adding New Knowledge to LLMs, Rapid Application Development with Large Language Models (LLMs)
- Multi-Agent Orchestration: Hands-on work with [Waldiez](#), a visual multi-agent orchestration framework for designing, running, and managing AI agent workflows.
- Advanced NLP with Deep Learning: End-to-end NLP projects covering LSTM-based toxicity detection (GloVe embeddings, class balancing), transformer fine-tuning (RoBERTa/BERT on STS-B, Pearson $r=0.89$), custom RAG pipeline with LLM-driven query expansion and hallucination mitigation, and DPO preference alignment of Llama-3.2-3B via LoRA.

ReGeneration Academy – 22nd Cycle by The Coca-Cola Company

Athens, GR (May 2026 – Jun. 2026)

- Selective employability program. Completed training in Business Readiness, Career Readiness, and AI Readiness (Presentation & Communication, Project Management, AI tools).

Master's in Health Statistics and Data Analytics (90 ECTS)

Thessaloniki, GR (2022 - 2024)

Aristotle University of Thessaloniki - English-taught program - Medicine / Electrical Engineering and Informatics

- Specialization: Meta-analysis, Environmental epidemiology.
- Thesis. Kopalidis, T., E. Samoli (2023). Effect of short-term exposure to ambient temperature on pediatric hospital admissions in Athens. <https://doi.org/10.26262/heal.auth.ir.353982>

Certification on Google Data Analytics

Online in Google

Thessaloniki, GR (2021)

Bachelor's in Physics (244 ECTS)

Aristotle University of Thessaloniki

Thessaloniki, GR (2020)

- Specialization: Medical Physics, Nuclear Physics.

PUBLICATIONS

1. Nikiforiadis, A.,..., **Kopalidis, T.**, et al. (2026) Transforming Urban Mobility Behavior through Gamification and Local Incentives in Thessaloniki, Greece (accepted).
2. Kopalidis, T., et al. (2024). Author of "Advances in facial expression recognition: a survey of methods, benchmarks, models, and datasets. *Information*, 15(3), 135". DOI: [10.3390/info15030135](https://doi.org/10.3390/info15030135) (Highly Cited Article - top100 most-cited authors in information)

CERTIFICATIONS

Intersection of AI and Environmental Sustainability - EY & Microsoft

Athens, GR Jun. (2026)

- Explored applications of AI in sustainable development and environmental impact reduction

The AI Engineer Course 2026: Complete AI Engineer Bootcamp

Thessaloniki, GR (2026)

- Complete AI Engineer Training: Python, NLP, Transformers, LLMs, LangChain, Hugging Face, APIs (30 hours) – Udemy

Microsoft Azure AI Essentials Professional Certificate

Thessaloniki, GR (2026)

Certification on PostGreSQL

Thessaloniki, GR (2026)

- The Complete SQL Bootcamp (Udemy): Go from Zero to Hero – Udemy in 2 months training seminar.

Certification on The Business Data Analyst

Thessaloniki, GR (2024)

- Business data analyst course 2025 (22 hours) – Udemy

Certification on Machine Learning

Coursera online courses London and Duke University

Thessaloniki, GR (2024)

- Developed my own machine learning project: collecting a dataset, training a model and testing it in 2-month seminar

ADDITIONAL INFORMATION

Languages Greek (Native), English (C2), French (B1)

Interests Marathon Athlete, World Travel, Read History Books, Gym, Coding, Sudoku

Military Service Fulfilled